Chapter one

Introduction to computer maintenance and preventive maintenance

1.0. What is Maintenance?

It is an operation by which *proactive measures* are taken to prevent computers from failure and a failed computer is made back to its normal working condition

When we talk about maintenance, our aims are to:

- **preserve** our IT systems in optimal condition
- **fix problems** that occur
- **Upgrade** the existing systems to minimise future risks to the business.

This will require maintenance of hardware, software and data.

Why Maintain?

Think of a computer as being like a human body. We are advised to perform certain tasks to keep our body in good shape. These maintenance tasks such as eating well, avoiding some harmful activities and getting exercise keep our body in good condition. If we don't maintain our bodies, we might end up in the hospital with problems that result from the abuse such as high blood pressure from eating too much salt, obesity from not exercising or lung problems from smoking. Similarly, if we allow our computers to intake harmful materials (such as dust) and programs (such as viruses) and we don't keep our software in good condition (updated) then the computer may also end up with significant problems that are then harder to deal with. Proper maintenance of a computer, on the other hand, can keep it running smoothly for years and years.

What do you maintain?

- you maintain Computer Hardware and Software
- Faulty parts of the computer

How do you maintain?

Using your Knowledge:

- ✓ Knowledge of Assembling & disassembling hardware parts
- ✓ Knowledge Of Identifying faulty components of the PC
- ✓ Knowledge of Installing and configuring hardware
- ✓ Knowledge of Installing and configuring software.

Using tools:

- ✓ Different types of screw drives (flat, star etc..)
- ✓ Operating system SW CD/DVD
- ✓ Application, Drivers, antivirus etc. Sw CD/DVD

1.1. Why Perform Computer Maintenance Procedures?

Maintaining your computer is not only good for your computer, but for you as well. Practicing good maintenance procedures will save you money (less chance of items breaking down and having to replace them), time (improved performance means less waiting time and less breaking down means more usage) and protects your data (by backing up the data, for example).

1.2. Types of maintenance

Maintenance falls into two broad categories:

- **Reactive** (or **non-routine)**.
- **preventative** (or **routine** maintenance), and

A. Reactive maintenance:

Reactive maintenance refers to actions taken to fix problems *after* they have occurred. To continue with the car example, when you get a flat tyre and have to replace it, this is a simple example of reactive maintenance. You can probably think of many IT examples. Replacing a broken cable is one.

Apart from preventative and reactive maintenance, there is another type of maintenance that deals with **upgrading** the organisation's **infrastructure** to minimise the level of risk to business continuity.

B. Preventative maintenance:

Preventive maintenance is a regular and systematic inspection, cleaning, and replacement of worn parts, materials, and systems. Preventive maintenance helps to prevent failure of parts, materials, and systems by ensuring that they are in good working order.

We need to carry out preventative maintenance on a periodic basis to *prevent* problems occurring in the future that may interrupt business continuity.

Why should you do preventive maintenance?

- Reduced the probability of hardware or
- software problems by systematically and
- periodically checking hardware and software
- to ensure proper operation

What are the benefits of preventive maintenance?

- Increases data protection
- Extends life of components
- Increases equipment stability
- Reduces repair costs
- Reduces number of equipment failures

Specific devices require different preventative maintenance procedures. However, there are a few broad areas that can be considered which require preventative maintenance practices. These include:

- **Protection of equipment due to changes in electrical supply**: Surges and 'brownouts' cause major damage to computer devices. Un-interruptible power supplies (UPS), power conditioners and surge protection devices are all valuable preventative maintenance tools for any computerised device.
- **Protection from environmental conditions**: Humidity, temperature variation and dust are major causes of computer device failures. Actions taken to limit these factors in the workplace are valuable preventative maintenance tasks.
- **Protection of data using backups**: It is vital that backups of both user data and system configurations are done regularly.
- **Protection of data from threats**: Data needs to be protected from viruses, malware, hackers and so on, through the use of both hardware and software security measures.
- **Keeping software updated through service packs, patches and driver upgrades**
- **Checking integrity and performance by using diagnostic tools**: Routine running of any in-built diagnostics and/or checking for display of maintenance messages generated by the device.

Protecting critical hardware:

Facility protection:

Strategies must be in place to protect IT equipment from water damage, fire, contamination, power failure and theft. Some of these strategies include:

- the implementation of **early warning systems** to detect water leaks, fire and air-borne contaminants
- **devices to continue power supply** should there be a power failure such as a UPS (uninterruptible power supply) as well as on-site power generation
- **security access** to computer facilities such as swipe cards or entry of a security number
- **Recording** serial numbers, asset numbers, and location and allocation details of Protection from environmental conditions

Protection of data — backup

All organisations need strategies in place to:

- Backup critical data, and Ensure that data backup is being undertaken according to organisational policies. This will involve both **server backup** and **workstation backup**.

As mentioned, staff in a client/server organisation is generally encouraged **not to store data** on their own hard drives. However, where an organisation's data is stored on a workstation hard drive, there must be some procedure in place for regularly backing it up.

Types of backup

An organisation will have policies that relate to:

- the **frequency** of backups (daily, weekly, monthly)

- the **time of day** backups are done
- **how long** backups are kept
- **Where** backups should be stored.

Also, there are different types of backup. You may not need to back up all files every time you back up. Different options include:

- backup of selected directories
- incremental backup – backup of only files that have been created or changed since the last full or incremental backup
- Differential backup – backup of files that have been created or changed since the last full backup.

Backup scheduling is an important part of any preventative maintenance plan. Windows provides a backup and restore tool, and this type of software is also provided by third parties.

Protection of data from threats:

Because of the widespread interconnectivity of computers today, and the potential for intrusion, theft, damage, and so on, organisations need to have clear policies and procedures to be followed to minimise these threats. Central to this will be a number of preventative maintenance procedures, as outlined below.

Service packs, patches and operating system updates

It has become obvious in recent years that when operating systems are released, they are not finished products. Because they are so complex, even after a period of rigorous testing, security flaws are often discovered after distribution. Anti-virus and other security tools cannot protect the system from operating system holes.

For critical security holes, as soon as the flaw is discovered, the software manufacturer quickly develops and releases a **patch,** which is a small software update to eliminate the hole. A group of patches is sometimes released as a major update, or **service pack**.

Microsoft makes these updates available for free.

Anti-malware software

Malware has increased in significance over recent years. Included in this category are:

- **Trojans**, which appear to be harmless programs, are actually designed to either do damage or carry out a range of malicious activities
- **Dialers**, programs that change the number you use for a dial-up modem, causing increases in charges
- **Worms**, generally spread as email attachments
- **Spyware, adware and browser hijackers** collect information from your computer or change the Internet options in the browser.

Anti-malware software helps to prevent a computer from these attacks. However, installing it and forgetting about it provides insufficient protection. Regular updates need to be carried out.

Antivirus software

Viruses have been around for a long while, and have been overtaken somewhat by other types of threat. However, using antivirus software which is regularly updated is still an important preventative maintenance measure for computer systems.

Firewalls

There are two types of firewalls — hardware and software firewalls. **Hardware firewalls** offer the best protection against intrusion, but they are expensive for small companies. Desktop or a **software firewall** is useful, but malicious software may find ways to bypass it.

Keeping software updated

Today's IT equipment is complex, and so is the software that supports it. Often a device is purchased with a projected life of several years, but in the mean time, other equipment and operating systems that it is used with are updated. Sometimes this means that a perfectly good piece of equipment no longer works, or works unsatisfactorily.

So new software for these devices (drivers) is developed and released. Keeping abreast of these various driver updates is also important in preventing problems before they arise.

Checking integrity and performance

Computer operating systems are generally provided with an array of diagnostic tools that can be used to check whether there are either hardware or software problems with the machine, or whether steps can be taken to improve its performance.

For example, Disk Defragmenter and Disk Cleanup are tools available in Windows operating systems. A whole range of other utilities is also available from other software manufacturers. Visit a website like download.com (http://www.download.com) and browse the range of software available.

Guidelines for developing a PC preventive maintenance plan

The following table gives suggestions for cleaning the parts of a computer.

Guidelines for developing a PC preventive maintenance plan

Component	Maintenance	How Often
Inside the case	• Make sure air vents are clear. • Use compressed air to blow the dust out of the case, or use a vacuum to clean vents, power supply, and fan. • Ensure that chips and expansion cards are firmly seated.	Yearly
CMOS setup	• Keep a backup record of setup (for example, using Norton Utilities or CMOS Save).	Whenever changes are made
Floppy drive	• Only clean the floppy drive when the drive does not work.	When the drive fails
Hard drive	• Perform regular backups. • Automatically execute a virus scan program at startup. • Defragment the drive and recover lost clusters regularly. • Don't allow smoking around the PC. • Place the PC where it will not be jarred, kicked, or bumped.	At least weekly At least daily Monthly Always Always

Guidelines for developing a PC preventive maintenance plan (continued)

Component	Maintenance	How Often
Keyboard	• Keep the keyboard clean. • Keep the keyboard away from liquids.	Monthly Always
Mouse	• Clean the mouse rollers and ball.	Monthly
Monitor	• Clean the screen with a soft cloth. • Make sure air vents are clear.	At least monthly Always
Printers	• Clean out the dust and bits of paper, using compressed air and a vacuum. Small pieces of paper can be removed with tweezers, preferably insulated ones. • Clean the paper and ribbon paths with a soft, lint-free cloth. • Don't re-ink ribbons or use recharged toner cartridges. • If the printer uses an ozone filter, replace it as recommended by the manufacturer. • Replace other components as recommended by the manufacturer.	At least monthly or as recommended by the manufacturer
UPS/ Suppressors	• Run weak battery test. • Run diagnostic test.	As recommended by manufacturer
Software	• If directed by your employer, check that only authorized software is present. • Regularly delete files from the Recycle Bin and \Temp directories.	At least monthly
Written record	• Keep a record of all software, including version numbers and the OS installed on the PC. • Keep a record of all hardware components installed, including hardware settings. • Record when and what preventive maintenance is performed. • Record any repairs done to the PC.	Whenever changes are made

Whole system maintenance:

Maintenance of a computer system will also involve:

- checking **system event logs** regularly
- viewing **POST** results
- routine checking using **system monitoring** utilities that track system temperatures, voltages and fan speeds
- checking for **dust** accumulation, particularly around fans and vents
- **updating drivers** for printers, modems, soundcards, video cards and so on, as needed
- **updating operating system** and **application software** with the latest service packs; eg later versions of Windows allow updates to be automatically downloaded and installed, but this can be disabled
- **Updating anti-virus software** and **virus definitions**.

Hard disk maintenance:

Software for carrying out hard disk maintenance is provided by the operating system utilities, as well as by third party software. Typical maintenance will include:

- **removal of unwanted files** — this can include old files, temporary files, downloaded files, corrupt files, Internet cookies, and browser history
- **removal** (uninstalling) **of unwanted software**
- **backup** — this may be by means of standard backup/restore software, or through imaging software such as Norton Ghost
- **cleaning up the registry**
- **defragmenting** files
- **creating system restore**/boot disks
- **scanning** for viruses, spyware, adware, malware, and so on
- **Disk checking** using standard diagnostic tests.

1.3. General concepts about PC and tech support

A *personal computer (PC)* is a computing device made up of many distinct electronic components that all function together in order to accomplish some useful task (such as adding up the numbers in a spreadsheet or helping you write a letter). By this definition, note that we're describing a computer as having many distinct parts that work together. Most computers today are modular. That is, they have components that can be removed and replaced with a component

of similar function in order to improve performance. Each component has a very specific function. Personal computer components, including the following:

a. Motherboards
b. Processors
c. Memory
d. Storage devices
e. Power supplies
f. Display devices
g. Input devices
h. Adapter cards
i. Ports and cables
j. Cooling systems

1.4. What is Technical support?

When you buy a computer, purchase software, or sign some type of agreement for a service (such as the Internet), you should get some form of support. Support is there to help you if things don't work as expected and to fix things when stuff invariably breaks.

Technical support is where you go when you have questions or problems with a product. Technical support may be acquired from manufacturer's Web site or you can find some excellent tech support freely available on the Web Windows Help and Support service is also the built-in help system for Windows.

- Talk staff or clients through a series of actions, either face-to-face or over the phone, to help set up systems or resolve issues
- Troubleshoot system and network problems, diagnosing and solving hardware or software faults
- Replace parts as required

In general, technical support provide help regarding specific problems with a product or service, rather than providing training, provision or customization of product, or other support services.

i. PC service technician

- Goes to a customer site in response to a service call
- Usually not responsible for ongoing maintenance but do interact with users
- Also known as computer repair technician or field service technician.

ii. PC support technician

- Works on site, works closely with users and is responsible for ongoing PC maintenance
- Prepare for a problem by performing routine preventative maintenance, keeping good records, and making backups of data






1.5. Basic Hardware Components

Hardware is the physical equipment needed for a computer to function properly. The basic hardware parts are briefly described here. Many computer lab managers will already have been exposed to computer hardware through other courses, but those desiring additional information can find it in a variety of texts and online sources. A desktop computer is used in the photos, but all of this equipment is also found (in a more compact arrangement) in a laptop computer.

a) **Case.** The computer case (also called a tower or housing) is the box that encloses many of the parts shown below. It has attachment points, slots and screws that allow these parts to be fitted onto the case. The case is also sometimes called the CPU, since it houses the CPU (central processing unit or processor), but this designation can lead to confusion. Please see the description of the processor, below.

b) **Power Supply.** The power supply is used to connect all of the parts of the computer described below to electrical power. It is usually is found at the back of the computer case.



c) **Fan.** A fan is needed to disperse the significant amount of heat that is generated by the electrically powered parts in a computer. It is important for preventing overheating of the various electronic components. Some computers will also have a heat sink (a piece of fluted metal) located near the processor to absorb heat from the processor.

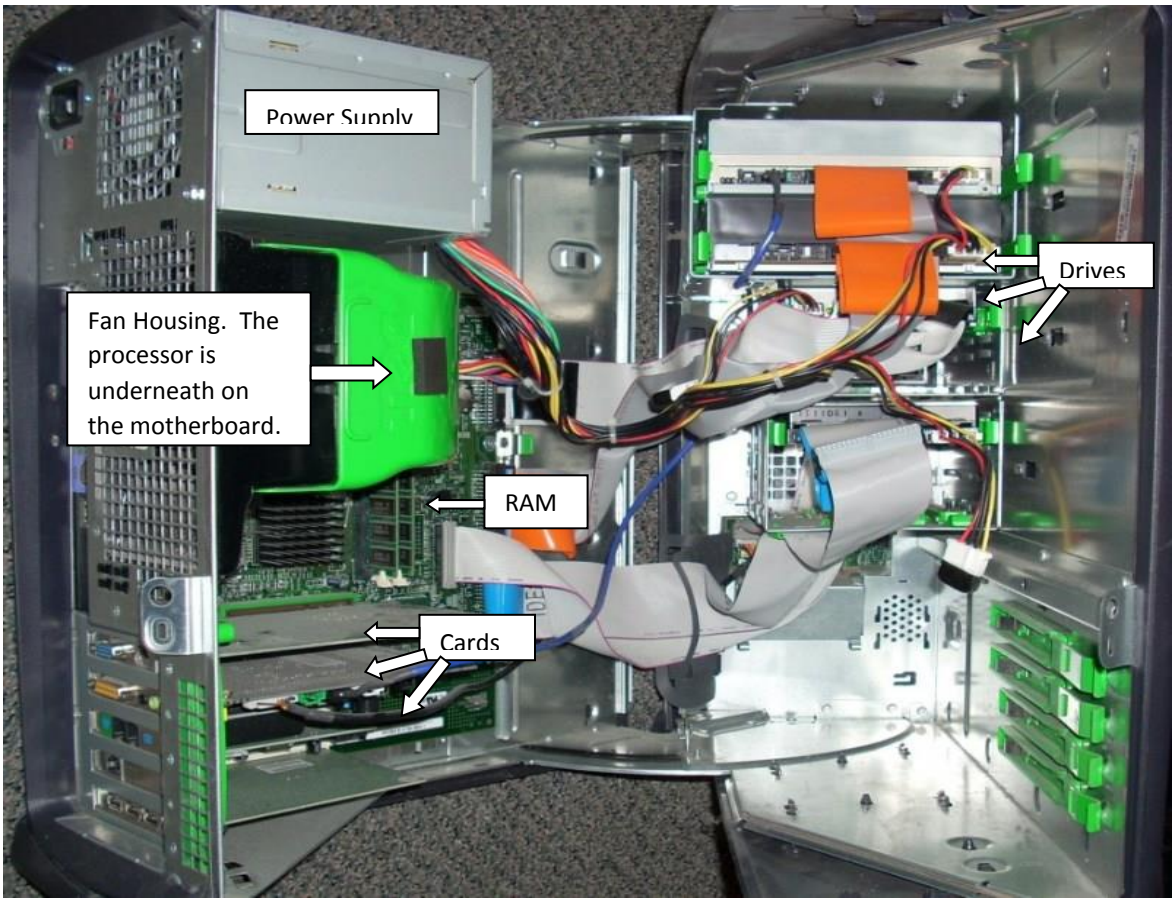
d) **Motherboard.** The motherboard is a large electronic board that is used to connect the power supply to various other electronic parts, and to hold these parts in place on the computer. The computer's memory (RAM, described below) and processor are attached to the motherboard. Also found on the motherboard is the BIOS (Basic Input and Output System) chip that is responsible for some fundamental operations of the computer, such as linking hardware and software. The motherboard also contains a small battery (that looks like a watch battery) and the chips that work with it to store the system time and some other computer settings.



e) **Drives.** A computer's drives are the devices used for long term storage of information. The main storage area for a computer is its internal **hard drive** (also called a hard disk). The computer should also have disk drives for some sort of removable storage media. A **floppy disk drive** was very common until recent years, and is still found on many older desk top computers. It was replaced by

f) **CD-ROM** and **DVD drives**, which have higher storage capacities. The current standard is a **DVD-RW** drive, which can both read and write information using both CD and DVD disks. The USB ports (described later) on a computer can also be used to connect other storage devices such as **flash drives** and **external hard drives**.





Inside a Desktop Computer Case

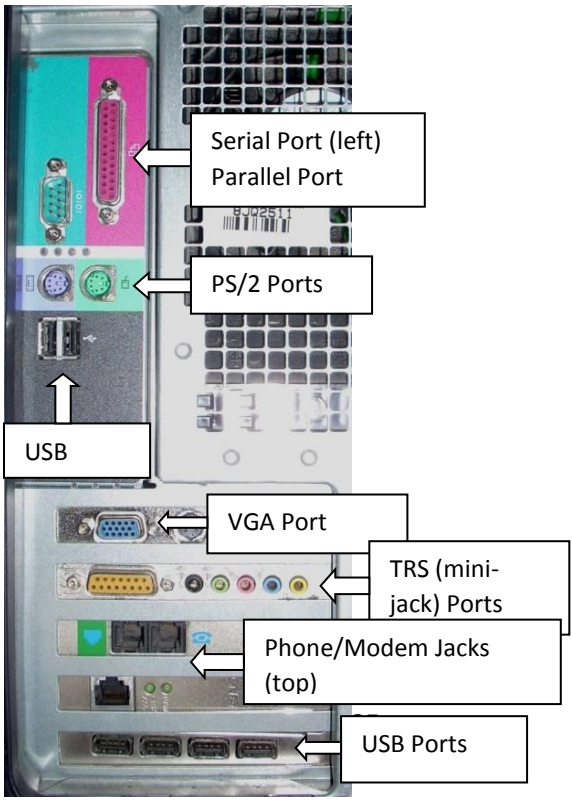
g) Cards. This term is used to describe important tools that allow your computer to connect and communicate with various input and output devices. The term "card" is used because these items are relatively flat in order to fit into the slots provided in the computer case. A computer will probably have a sound card, a video card, a network card and a modem.

h) RAM. RAM is the abbreviation for random access memory. This is the short term memory that is used to store documents while they are being processed. The amount of RAM in a computer is one of the factors that affect the speed of a computer. RAM attaches to the motherboard via some specific slots. It is important to have the right type of RAM for a specific computer, as RAM has changed over the years.

i) Processor. The processor is the main "brain" of a computer system. It performs all of the instructions and calculations that are needed and manages the flow of information through a computer. It is also called the CPU (central processing unit), although this term can also be used to describe a computer case along with all of the hardware found inside it. Another name for the processor is a computer "chip" although this term can refer to other lesser processors (such as the BIOS). Processors are continually evolving and becoming faster and more powerful. The speed of a processor is measured in megahertz (MHz) or gigahertz (GHz). An older computer might have a processor with a speed of 1000 MHz (equivalent to 1 GHz) or lower, but processors with speeds of over 2 GHz are now common. One processor company, Intel, made a popular series of processors called Pentium. Many reconditioned computers contain Pentium II, Pentium III and Pentium 4 processors, with Pentium 4 being the fastest of these.



j) Peripheral hardware. Peripheral hardware is the name for the computer components that are not found within the computer case. This includes input devices such as a **mouse, microphone** and **keyboard**, which carry information from the computer user to the processor, and output devices such as a **monitor, printer** and **speakers**, which display or transmit information from the computer back to the user**.**



Back of Desktop Computer Showing Ports

1.6. Computer Ports

The peripheral hardware mentioned above must attach to the computer so that it can transmit information from the user to the computer (or vice versa). There are a variety of ports present on a computer for these attachments. These ports have gradually changed over time as computers have changed to become faster and easier to work with. Ports also vary with the type of equipment that connects to the ports. A computer lab manager should become familiar with the most common ports (and their uses), as described below.

a) **Serial Port.** This port for use with 9 pin connectors is no longer commonly used, but is found on many older computers. It was used for printers, mice, modems and a variety of other digital devices.
b) **Parallel Port.** This long and slender port is also no longer commonly used, but was the most common way of attaching a printer to a computer until the introduction of USB ports (see below). The most common parallel port has holes for 25 pins, but other models were also manufactured.
c) **VGA.** The Video Graphics Array port is found on most computers today and is used to connect video display devices such as monitors and projectors. It has three rows of holes, for a 15 pin connector.
d) **PS/2**. Until recently, this type of port was commonly used to connect keyboards and mice to computers. Most desktop computers have two of these round ports for six pin connectors, one for the mouse and one for the keyboard.
e) **USB.** The Universal Serial Bus is now the most common type of port on a computer. It was developed in the late 1990s as a way to replace the variety of ports described above. It can be used to connect mice, keyboards, printers, and external storage devices such as DVD-RW drives and flash drives. It has gone through three different models (USB 1.0, USB 2.0 and USB 3.0), with USB 3.0 being the fastest at sending and receiving information. Older USB devices can be used in newer model USB ports.
f) **TRS.** TRS (tip, ring and sleeve) ports are also known as ports for mini-jacks or audio jacks. They are commonly used to connect audio devices such as headphones and microphones to computers.

g) **Ethernet.** This port, which looks like a slightly wider version of a port for a phone jack, is used to network computers via category 5 (CAT5) network cable. Although many computers now connect wirelessly, this port is still the standard for wired networked computers. Some computers also have the narrower port for an actual phone jack. These are used for modem connections over telephone lines.

1.7. Lab procedures and maintenance tools

a) General Safety

Safe working conditions help prevent injury to people and damage to computer equipment. A safe workspace is clean, organized, and properly lighted. Everyone must understand and follow safety procedures.

Follow the basic safety guidelines to prevent cuts, burns, electrical shock, and damage to eyesight.

As a best practice, make sure that a fire extinguisher and first-aid kit are available in case of fire or injury. Poorly placed or unsecured cables can cause tripping hazards in a network installation. Cables should be installed in conduit or cable trays to prevent hazards.

This is a partial list of basic safety precautions to use when working on a computer:

- ☞ Remove your watch and jewelry and secure loose clothing.
- ☞ Turn off the power and unplug equipment before performing service.
- ☞ Cover sharp edges inside the computer case with tape.
- ☞ Never open a power supply or a CRT monitors.
- ☞ Do not touch areas in printers that are hot or that use high voltage.
- ☞ Know where the fire extinguisher is located and how to use it.
- ☞ Keep food and drinks out of your workspace.
- ☞ Keep your workspace clean and free of clutter.
- ☞ Bend your knees when lifting heavy objects to avoid injuring your back.

1.8. Static energy and its effect on computers

- **Static energy or** Electrostatic Discharge (ESD) which is more commonly referred to as static electricity is probably the greatest problem when a user is unwrapping newly purchased computer parts and components while preparing to assemble the computer.
- The human body builds up static electricity, which can damage components such as circuits and integrated chips.
- To prevent damage from ESD, make sure **to use** an antistatic mat or wrist strap, and touch unpainted metal frequently to ground yourself before you open the case and reach in there.
- Static electricity is created when electrons move back and forth between atoms.
- Electrostatic Discharge (ESD) is the term used to describe the transfer of static electricity from one object to another.
- Static electricity is the buildup of an electric charge resting on a surface. ESD occurs when this buildup jumps to a component and causes damage. ESD can be destructive to the electronics in a computer system.
- At least 3,000 volts of static electricity must build up before a person can feel ESD. For example, static electricity can build up on you as you walk across a carpeted floor. When you touch another person, you both receive a shock. If the discharge causes pain or makes a noise, the charge was probably above 10,000 volts. By comparison, less than 30 volts of static electricity can damage a computer component.

ESD can cause permanent damage to electrical components. Follow these recommendations to help prevent ESD damage:

- ☞ Keep all components in antistatic bags until you are ready to install them.
- ☞ Use grounded mats on workbenches.
- ☞ Use grounded floor mats in work areas.
- ☞ Use antistatic wrist straps when working on computers.

Climate affects computer equipment in a variety of ways:

- ☞ If the environment temperature is too high, equipment can overheat.
- ☞ If the humidity level is too low, the chance of ESD increases.
- ☞ If the humidity level is too high, equipment can suffer from moisture damage.

1.9. Safety rules

Whenever you work with computers, safety is not compromised. Keep these important tips in mind as you work inside your computer:

i. **Remember to Flip the Switch-** Always remember to turn the power off before servicing anything. This should always be your first step. Do not even open the computer case and not move or install while the computer is on.

ii. **Unplug for Extra Safety**-As a second precaution, it is wise to unplug the computer from the wall or power strip. If there was any doubt as to whether the computer was off before, it's settled now.

iii. **Avoid Smoke and Smells**-See smoke coming from the power supply or inside the case or smell a burning or solder scent? If so:

- Stop what you're doing immediately.
- Unplug the computer from the wall.
- Allow the PC to cool or discharge unplugged for at least 5 minutes.
- Finally, if you know which device was generating the smoke or smell, remove and replace it as soon as you can. Don't try to repair a device that's been damaged to this extent, especially if it's a power supply.

IV. **Remove Hand Jewelry**

- An easy way to get electrocuted is to work around a high voltage device like a power supply with metal rings, watches, or bracelets on.
- Remove anything conductive from your hands before working inside your computer, especially if you're doing something like testing your power supply.

V. **Avoid Capacitors**

Capacitors are miniature electronic components contained in many of the parts inside a PC. Capacitors can store electric charge for a short while after the power is turned off so it's a wise decision to wait a few minutes after pulling the plug before working on your PC.

VI. Do not allow **food and drinks** in the work area

Keep computer disks away from magnetic fields, heat and cold.

Use an anti-static mat and grounding wrist strap.

Use anti-static bags to store and move computer components. Do not put more than one component in each bag. Components may be damaged or broken.

VII. Keep the work area clean and orderly

1.10. Hardware Tools

General Tool Use

For every job there is the right tool. Make sure that you are familiar with the correct use of each tool and that the correct tool is used for the current task. Skilled use of tools and software makes the job less difficult and ensures that tasks are performed properly and safely.



A toolkit should contain all the tools necessary to complete hardware repairs. As you gain experience, you learn which tools to have available for different types of jobs. Hardware tools are grouped into four categories:

- ESD tools
- Hand tools
- Cleaning tools
- Diagnostic tools

a) ESD Tools

There are two ESD tools: the antistatic wrist strap and the antistatic mat. The antistatic wrist strap protects computer equipment when grounded to a computer chassis. The antistatic mat protects computer equipment by preventing static electricity from accumulating on the hardware or on the technician.





Antistatic mat

b) <u>Hand Tools</u>

Most tools used in the computer assembly process are small hand tools. They are available individually or as part of a computer repair toolkit. Toolkits range widely in size, quality, and price. Common hand tools include the following:

- **Flat-head screwdriver**: Used to tighten or loosen slotted screws
- **Phillips-head screwdriver**: Used to tighten or loosen cross-headed screws
- **Torx screwdriver**: Used to tighten or loosen screws that have a star-like depression on the top, a feature that is mainly found on laptops
- **Hex driver**: Used to tighten or loosen nuts in the same way that a screwdriver tightens or loosens screws (sometimes called a nut driver)
- **Needle-nose pliers**: Used to hold small parts
- **Wire cutters**: Used to strip and cut wires
- **Tweezers**: Used to manipulate small parts
- **Part retriever**: Used to retrieve parts from locations that are too small for your hand to fit
- **Flashlight**: Used to light up areas that you cannot see well

c) Cleaning Tools

Having the appropriate cleaning tools is essential when maintaining and repairing computers. Using the appropriate cleaning tools helps ensure that computer components are not damaged during cleaning. Cleaning tools include the following:

- **Soft cloth**: Used to clean different computer components without scratching or leaving Debris
- **Compressed air**: Used to blow away dust and debris from different computer parts without touching the components
- **Cable ties**: Used to bundle cables neatly inside and outside of a computer
- **Parts organizer**: Used to hold screws, jumpers, fasteners, and other small parts and prevent them from getting mixed together

d) Diagnostic Tools

- **Digital Multimeter**

A digital multimeter is a device that can take many types of measurements. It tests the integrity of circuits and the quality of electricity in computer components. A digital multimeter displays the information on an LCD or LED.

- **Loopback Adapter**

A loopback adapter also called a loopback plug, tests the basic functionality of computer ports. The adapter is specific to the port that you want to test.

Check the condition of cables, components, and peripherals. Clean components to reduce the likelihood of overheating. Repair or replace any components that show signs of damage or excessive wear.

Use c following tasks as a guide to create a hardware maintenance program.

- Remove dust from fan intakes.
- Remove dust from the power supply.
- Remove dust from components inside the computer.
- Clean the mouse and keyboard.
- Check and secure loose cables.



1.11. Software Tools

Protection Software Tools

Each year, viruses, spyware, and other types of malicious attacks infect millions of computers. These attacks can damage operating systems, applications, and data. Computers that have been infected may even have problems with hardware performance or component failure.

To protect data and the integrity of the operating system and hardware, use software designed to guard against attacks and to remove malicious programs.

Various types of software protect hardware and data. The figure gives more information on these protection software tools:

- **Windows 7 Action Center** - Checks the status of essential security settings. The Action Center continuously checks to make sure that the software firewall and antivirus programs are running. It also ensures that automatic updates download and install automatically.
- **Antivirus program** - Protects against virus attacks.
- **Antispyware program** - Protects against software that sends information about web surfing habits to an attacker. Spyware can be installed without the knowledge or consent of the user.
- **Window 7 Firewall** - Runs continuously to protect against unauthorized communications to and from your computer.

Verify that installed software is current. Follow the policies of the organization when installing security updates, operating system updates, and program updates. Many organizations do not allow updates until extensive testing has been completed. This testing is done to confirm that the update will not cause problems with the operating system and software.

Use the tasks listed as a guide to create a software maintenance schedule that fits the needs of your computer equipment:

- Review security updates.
- Review software updates.
- Review driver updates.
- Update virus definition files.
- Scan for viruses and spyware.
- Remove unwanted programs
- Scan hard drives for errors.
- Defragment hard drives.

1.12. PREVENTIVE MAINTENANCE AND TROUBLESHOOTING

Preventive maintenance is a regular and systematic inspection, cleaning, and replacement of worn parts, materials, and systems. Preventive maintenance helps to prevent failure of parts, materials, and systems by ensuring that they are in good working order.

- Preventive maintenance is performed for the purpose of preventing problems before they occur.
- It is the care and servicing by personnel for the purpose of maintaining equipment and facilities in satisfactory operating condition by providing for systematic inspection, detection, and correction of incipient failures either before they occur or before they develop into major defects.
- The ideal preventive maintenance program would prevent all equipment failure before it occurs.

Be proactive in computer equipment maintenance and data protection. By performing regular maintenance routines, you can reduce potential hardware and software problems. Regular maintenance routines reduce computer downtime and repair costs.

A preventive maintenance plan is developed based on the needs of the equipment. A computer exposed to a dusty environment, such as a construction site, needs more attention than equipment in an office environment. High-traffic networks, such as a school network, might require additional scanning and removal of malicious software or unwanted files. Document the routine

maintenance tasks that must be performed on the computer equipment and the frequency of each task. This list of tasks can then be used to create a maintenance program.

Preventive Maintenance can be carried on the following computer parts.

i. System Unit

Your PC system unit has a lot of stuff inside—everything from memory chips to disk drives to power supplies. Check out these maintenance tips to keep your system unit from flaking out on you:

- Position your system unit in a clean, dust-free environment. Keep it away from direct sunlight and strong magnetic fields. In addition, make sure your system unit and your monitor have plenty of air flow around them, to keep them from overheating.
- Hook your system unit up to a surge suppressor to avoid deadly power spikes.
- Avoid turning your system unit on and off too often; it's better to leave it on all the time than incur frequent "power on" stress to all those delicate components. However…
- Turn off your system unit if you're going to be away for an extended period—anything longer than a day or two.
- Check all your cable connections periodically. Make sure all the connectors are firmly connected, and all the screws properly screwed—and make sure your cables aren't stretched too tight, or bent in ways that could damage the wires inside.
- If you're really adventurous, open up the system case periodically and vacuum or wipe the dust from the inside. (Just make sure the system unit is unplugged at the time!) Using a can of "compressed air" is also a good way to blast the dirt out of your system. You should also "dust" the inside of your CD-ROM drive, and even use a swab or commercial cleaner to clean the laser lens.

ii. Keyboard

Even something as simple as your keyboard requires a little preventive maintenance from time to time. Check out these tips:

- Keep your keyboard away from young children and pets—they can get dirt and hair and Silly Putty™ all over the place, and have a tendency to put way too much pressure on the keys.
- Keep your keyboard away from dust, dirt, smoke, direct sunlight, and other harmful environmental stuff. You might even consider putting a dust cover on your keyboard when it's not in use.
- Use a small vacuum cleaner to periodically sweep the dirt from your keyboard. (Alternately, you can use compressed air to blow the dirt away.) Use a cotton swab or

soft cloth to clean between the keys. If necessary, remove the keycaps to clean the switches underneath.

- If you spill something on your keyboard, disconnect it immediately and wipe up the spill. Use a soft cloth to get between the keys; if necessary, use a screwdriver to pop off the keycaps and wipe up any seepage underneath. Let the keyboard dry thoroughly before trying to use it again.

iii. Mouse

If you're a heavy Windows user, you probably put thousands of miles a year on your mouse. Just like a car tire, anything turning over that often needs a little tender loving care. Check out these mouse maintenance tips:

- Periodically open up the bottom of your mouse and remove the roller ball. Wash the ball with water (or perhaps a mild detergent). Use a soft cloth to dry the ball before reinserting it.
- While your mouse ball is removed, use compressed air or a cotton swab to clean dust and dirt from the inside of your mouse. (In extreme cases, you might need to use tweezers to pull lint and hair out of your mouse—or use a small knife to scrape packed crud from the rollers.)
- Always use a mouse pad—they really do help keep things rolling smoothly, plus they give you good traction. (And while you're at it, don't forget to clean your mouse pad with a little spray cleaner—it can get dirty, too.)

iv. Monitor

If you think of your monitor as a little television set, you're on the right track. Just treat your monitor as you do your TV and you'll be okay. That said, look at these preventive maintenance tips:

- As with all other important system components, keep your monitor away from direct sunlight, dust, and smoke. Make sure it has plenty of ventilation, especially around the back; don't cover the rear cooling vents with paper or any other object, and don't set anything bigger than a small plush toy on top of the cabinet.
- Don't place any strong magnets in close proximity to your monitor. (This includes external speakers!)
- With your monitor turned off, periodically clean the monitor screen. Spray standard glass cleaner on a soft cloth (anti-static type, if possible) and then wipe the screen clean.
- Don't forget to adjust the brightness and contrast controls on your monitor every now and then. Any controls can get out of whack—plus, your monitor's performance will change as it ages, and simple adjustments can often keep it looking as good as new.

v. Printer

Your printer is a complex device with a lot of moving parts. Follow these tips to keep your printouts in good shape:

- Use a soft cloth, mini-vacuum cleaner, and/or compressed air to clean the inside and outside of your printer on a periodic basis. In particular, make sure you clean the paper path of all paper shavings and dust.
- If you have an ink-jet printer, periodically clean the ink jets. Run your printer's cartridge cleaning utility, or use a small pin to make sure they don't get clogged.
- If you have a laser printer, replace the toner cartridge as needed. When you replace the cartridge, remember to clean the printer cleaning bar and other related parts.
- Don't use alcohol or other solvents to clean any rubber or plastic parts—you'll do more harm than good!

1.1. Basic Aspects of Troubleshooting

Troubleshooting is a systematic approach to locating the cause of a fault in a computer system. A good preventive maintenance program helps minimize failures. With fewer failures, there is less troubleshooting to do, thus saving an organization time and money. Preventive maintenance can also include upgrading certain hardware or software such as a hard drive that is making noise, upgrading memory that is insufficient, or installing software updates for security or reliability.

Troubleshooting is a systematic approach to finding the cause of a computer related problem. By following basic troubleshooting rules you can isolate problem areas with ease. Troubleshooting is needed to develop and maintain complex systems where the symptoms of a problem can have many possible causes. Troubleshooting requires identification of the malfunction(s) or symptoms within a system. Then, experience is commonly used to generate possible causes of the symptoms. Determining the most likely cause is a process of elimination - eliminating potential causes of a problem. Finally, troubleshooting requires confirmation that the solution restores the product or process to its working state.

1.1.1. Basic Diagnostic Procedures

Just as all artists have their own style, all technicians have their own way to troubleshoot. Some people use their instincts; others rely on advice from other people. The most common

troubleshooting tips can be condensed into a step-by-step process. You try each step, in order. If the first step doesn't narrow down the problem, you move on to the next step.

In this section, we'll look at each step in the troubleshooting process.

Step 1: Define the Problem

If you can't define the problem, you can't begin to solve it. You can define the problem by asking questions of the user. Here are a few questions to ask the user to aid in determining what the problem is, exactly:

Can you show me the problem? This question is one of the best. It allows the user to show you exactly where and when he experiences the problem.

How often does this happen? This question establishes whether this problem is a one-time occurrence that can be solved with a reboot or whether a specific sequence of events causes the problem to happen. The latter usually indicates a more serious problem that may require software installation or hardware replacement.

Has any new hardware been installed recently? New hardware can mean compatibility problems with existing devices. Some Plug and Play devices install with the same resource settings as an existing device. This can cause both devices to become disabled.

Have any other changes been made to the computer recently? If the answer is yes, ask if the user can remember approximately when the change was made. Then ask her approximately when the problem started. If the two dates seem related, then there's a good chance that the problem is related to the change. If it's a new hardware component, check to see that the hardware component was installed correctly.

Step 2: Check the Simple Stuff First

This step is the one that most experienced technicians overlook. Often, computer problems are the result of something simple. Technicians overlook these problems because they're so simple that the technicians assume they *couldn't* be the problem. Some examples of simple problems are shown here:

Is it plugged in? And plugged in at both ends? Cables must be plugged in at *both ends* in order to function correctly. Cables can easily be tripped over and inadvertently pulled from their sockets.

Is it turned on? This one seems the most obvious, but we've all fallen victim to it at one point or another. Computers and their peripherals must be turned on in order to function. Most have power switches with LEDs that glow when the power is turned on.

Is the system ready? Computers must be ready before they can be used. *Ready* means the system is ready to accept commands from the user. An indication that a computer is ready is when the operating system screens come up and the computer presents you with a menu or a command prompt. If that computer uses a graphical interface, the computer is ready when the mouse pointer appears. Printers are ready when the Online or Ready light on the front panel is lit.

Do the chips and cables need to be reseated? You can solve some of the strangest problems (random hang-ups or errors) by opening the case and pressing down on each socketed chip. This remedies the chip-creep problem discussed later in this chapter. In addition, you should reseat any cables to make sure that they're making good contact.

Step 3: Check to See If It's User Error

This error is common but preventable. The indication that a problem is due to user error is when a user says he can't perform some very common computer task, such as printing or saving a file. As soon you hear these words, you should begin asking questions to determine if it's simply a matter of teaching the user the correct procedure. A good question to ask following his statement of the problem is, "Were you *ever* able to perform that task?" If he answers no to this question, it means he's probably doing the procedure wrong. If he answers yes, you must move on to another set of questions.

Step 4: Restart the Computer

It's amazing how often a simple computer restart can solve a problem. Restarting the computer clears the memory and starts the computer with a clean slate. Whenever we perform phone support, we always ask the customer to restart the computer and try again. If restarting doesn't work, try powering down the system completely and then powering it up again (rebooting). More often than not, that will solve the problem.

Step 5: Determine If the Problem Is Hardware or Software Related

This step is important because it determines on what part of the computer you should focus your troubleshooting skills. Each part requires different skills and different tools.

To determine whether a problem is hardware or software related, you can do a few things to narrow down the issue. For instance, does the problem manifest itself when the user uses a

particular piece of hardware (a modem, for example)? If it does, the problem is more than likely hardware related.

This step relies on personal experience more than any of the other steps do. You'll without a doubt run into strange software problems. Each one has a particular solution. Some may even require reinstallation of the software or the entire operating system.

Step 6: If the Problem Is Hardware Related, Determine Which Component Is Failing

Hardware problems are pretty easy to figure out. If the modem doesn't work, and you know it isn't a software problem, the modem is probably the piece of hardware that needs to be replaced. With some of the newer computers, several components are integrated into the motherboard.

If you troubleshoot the computer and find a hardware component to be bad, there's a good chance that the bad component is integrated into the motherboard (for example, the parallel port circuitry) and the whole motherboard must be replaced—an expensive proposition, to be sure.

Step 7: Check Service Information Sources

As you may (or may not) have figured out by now, we're fond of old sayings. Another old saying applies here: "If all else fails, read the instructions." The service manuals are your instructions for troubleshooting and service information. Almost every computer and peripheral made today has service documentation in the form of books, service CD-ROMs, and websites.

The latter of the three is growing in popularity as more and more service centers get connections to the Internet.

Step 8: If It Ain't Broke …

When doctors take the Hippocratic oath, they promise to not make their patients any sicker than they already were. Technicians should take a similar oath. It all boils down to "If it ain't broke, don't fix it." When you troubleshoot, make one change at a time. If the change doesn't solve the problem, revert the computer to its previous state before making a different change.

Step 9: Ask for Help

If you don't know the answer, ask one of your fellow technicians. They may have run across the problem you're having and know the solution.

This solution does involve a little humility. You must admit that you don't know the answer. It's said that the beginning of wisdom is "I don't know." If you ask questions, you'll get answers, and you'll learn from the answers. Making mistakes is valuable as well, as long as you learn from them.

1.13. Recognizing and Isolating Issues

Your value as a technician increases as you gain experience because of the reduced time it takes you to accomplish common repairs. Your ability to troubleshoot by past experiences and gut feelings will make you more efficient and more valuable, which in turn will allow you to advance and earn a better income. This section will give you some guidelines you can use to evaluate common hardware issues that you're sure to face.

✓ POST Routines

Every computer has a diagnostic program built into its BIOS called the *power-on self-test* (*POST*). When you turn on the computer, it executes this set of diagnostics. Many steps are involved the POST, but they happen very quickly, they're invisible to the user, and they vary among BIOS versions. The steps include checking the CPU, checking the RAM, checking for the presence of a video card, and so on. The main reason to be aware of the POST's existence is that if it encounters a problem, the boot process stops. Being able to determine at what point the problem occurred can help you troubleshoot.

One way to determine the source of a problem is to listen for a *beep code*. This is a series of beeps from the computer's speaker. The number, duration, and pattern of the beeps can sometimes tell you what component is causing the problem. However, the beeps differ depending on the BIOS manufacturer and version, so you must look up the beep code in a chart for your particular BIOS. Different BIOS manufacturers use the beeping differently. AMI BIOS, for example, relies on a raw number of beeps and uses patterns of short and long beeps.

Another way to determine a problem during the POST routine is to use a *POST card*. This is a circuit board that fits into an ISA or PCI expansion slot in the motherboard and reports numeric codes as the boot process progresses. Each of those codes corresponds to a particular component being checked. If the POST card stops at a certain number, you can look up that number in the manual that came with the card to determine the problem.

Applying Basic Troubleshooting Techniques

Being able to identify and address basic troubleshooting issues is an important aspect of any PC technician's skill set. In the following sections, we will discuss the basic items to check for common problems.

✓ Identifying Motherboard and CPU Problems

Most motherboard and CPU problems manifest themselves by the system appearing completely dead. However, "completely dead" can be a symptom of a wide variety of problems, not only with the CPU or motherboard but also with the RAM or the power supply. So a POST card (described in the preceding section) may be helpful in narrowing down the exact component that is faulty.

When a motherboard fails, it's usually because it has been damaged. Most technicians can't repair motherboard damage; the motherboard must be replaced. Motherboards can become damaged due to physical trauma, exposure to electrostatic discharge (ESD), or short-circuiting. To minimize the risk of these damages, observe the following rules:

- Handle a motherboard as little as possible, and keep it in an antistatic bag whenever it's removed from the PC case.
- Keep all liquids well away from the motherboard, because water can cause a short circuit.
- Wear an antistatic wrist strap when handling or touching a motherboard.
- When installing a motherboard in a case, make sure you use brass stand-offs with paper washers to prevent any stray solder around the screw holes from causing a short circuit with the metal of the screw.

A CPU may fail because of physical trauma or short-circuiting, but the most common cause for a CPU not to work is failure to install it properly. With a PGA-style CPU, ensure that the CPU is oriented correctly in the socket. With an SECC-style CPU, make sure the CPU is completely inserted into its slot.

✓ **Identifying I/O Ports and Cables**

I/O ports include legacy parallel and serial, USB, and FireWire ports, all of which are used to connect external peripherals to the motherboard.

When a port doesn't appear to be functioning, check the following:

- The cables are snugly connected.
- The port has not been disabled in BIOS Setup.
- The port has not been disabled in Device Manager in Windows.
- No pins are broken or bent on the male end of the port or on the cable being plugged into it.

If you suspect that the cable, rather than the port, may be the problem, swap out the cable with a known good one. If you don't have an extra cable, you can test the existing cable with a multi-

meter by setting it to ohms and checking the resistance between one end of the cable and the other.

Use a pin-out diagram, if available, to determine which pin matches up to which at the other end. There is often—but not always—an inverse relationship between the ends. In other words, at one end pin 1 is at the left, and at the other end it's at the right on the same row of pins.

- ✓ **Identifying Cooling Issues**

A PC that works for a few minutes and then locks up is probably experiencing overheating because of a heat sink or fan not functioning properly. To troubleshoot overheating, first check all fans inside the PC to ensure they're operating, and make sure any heat sinks are firmly attached to their chips.

In a properly designed, properly assembled PC case, air flows in a specific path from the power supply fan through the vent holes. Cases are designed to cool by making the air flow in a certain way. Therefore, operating a PC with the cover removed can make a PC more susceptible to overheating, even though it's "getting more air."

Similarly, operating a PC with empty expansion slot backplates removed can inhibit a PC's ability to cool itself properly because the extra holes change the airflow pattern from what was intended by its design.

Although CPUs are the most common component to overheat, occasionally chips on other devices, particularly video cards, may also overheat. Extra heat sinks or fans may be installed to cool these chips.

- ✓ **Identifying Case Issues**

A PC case holds the drives in its bays, holds the power supply, and has lights and buttons on the front. For the first two of those functions, make sure that the drives and the power supply are tightly fastened in the case with screws.

If one of the lights or buttons on the front of the PC isn't functioning, remove the cover and check the wires that run from the back of that button/light to the motherboard. If the wire has become detached, reattach it. Refer to the motherboard manual or the writing on the motherboard itself to determine what goes where.

- ✓ **Identifying Hard-Disk System Problems**

Hard-disk system problems usually stem from one of three causes:

- The adapter (that is, the IDE or SCSI interface) is bad.

- The disk is bad.
- The adapter and disk are connected incorrectly.

The first and last causes are easy to identify, because in either case the symptom will be obvious: The drive won't work. You won't be able to get the computer to communicate with the disk drive.

However, if the problem is a bad disk drive, the symptoms aren't as obvious. As long as the BIOS POST routines can communicate with the disk drive, they're usually satisfied. But the POST routines may not uncover problems related to storing information. Even with healthy POST results, you may find that you're permitted to save information to a bad disk, but when you try to read it back, you get errors. Or the computer may not boot as quickly as it used to, because the disk drive can't read the boot information successfully every time.

In some cases, reformatting the drive can solve the problems described in the preceding paragraph. In other cases, reformatting brings the drive back to life only for a short while. The bottom line is that read and write problems usually indicate that the drive is malfunctioning and should be replaced soon.

✓ **Identifying Modem Problems**

The most common peripheral problems are those related to modem communications. The symptoms of these problems include the following:

- The modem won't dial.
- The modem keeps hanging up in the middle of the communications session.
- The modem spits out strange characters to the terminal screen.

If the modem won't dial, first check the simple things (for example, which the modem cable is plugged into the correct RJ-11 port on the modem and plugged into a working RJ-11 in the wall). Next, verify that it has been configured correctly in Windows, including its resource assignments.

Modem and port settings in Windows XP/2000 can be set through Control Panel _ Phone and Modems Options _ Modems tab, as shown in Figure 2.14.

Some modems work only under Windows because some of their functions rely on Windows software; these are called *Winmodems* or *software modems*. If such a modem doesn't work immediately upon installation, try running the Setup software that came with the modem.

If the configuration is correct, and Windows recognizes the modem, it should work for dial-up networking connections.

- ✓ **Identifying Keyboard and Mouse Problems**

Usually, keyboard problems are environmental. Keyboards get dirty, and the keys start to stick One way to clean a keyboard is with the keyboard cleaner sold by electronics supply stores. This cleaner foams up quickly and doesn't leave a residue behind. Spray it liberally on the keyboard and keys. Work the cleaner in between the keys with a stiff toothbrush. Blow away the excess with a strong blast of compressed air. Repeat until the keyboard functions properly. If you have to clean a keyboard that's had a soft drink spilled on it, remove the key caps before you perform the cleaning procedure; doing so makes it easier to reach the sticky plungers.

Similarly, most mouse problems, such as the pointer failing to move in one direction or the other or the pointer jumping around onscreen, are due to dirt building up inside the mouse.

To clean a standard mouse, remove the plate on the bottom of the mouse that holds the ball in place; then remove the ball and clean the inside chamber with an alcohol-dipped cotton swab. Clean the ball itself with mild soap and water. Don't use alcohol on the ball, because it tends to dry out the rubber.

- ✓ **Identifying Display Device Problems**

As a general rule, there are two types of video problems: no video and bad video. If there is no video, you should first check the monitor by transferring it to another machine that you know is working and see if it works there. If it does not work, then you should replace it (remember: not working on a monitor because of the electrical charge stored within). If the display does work once transferred to another machine, then you should focus on the video card: is it seated? Does it need replaced? etc.

If there is a display, but it is bad, then you should first focus on the settings. Make certain you have the correct driver for the monitor and that you are using settings that are appropriate for that monitor. If the problem persists after you have ruled out those possibilities, then you should focus on the video card and consider replacing it.

- ✓ **Identifying Sound Card Problems**

Sound cards are traditionally one of the most problem-ridden components in a PC. They demand a lot of PC resources and are notorious for being inflexible in their configuration. The most

common problems related to sound cards involve resource conflicts (IRQ, DMA, or I/O address). The problem is much less pronounced on PCI than on ISA cards.

Luckily, most sound-card vendors are aware of the problems and ship very good diagnostic utilities to help resolve them. Use your PC troubleshooting skills to determine the conflict, and then reconfigure until you find an acceptable set of resources that aren't in use. Some sound cards aren't completely Plug and Play compatible. Windows may detect that new hardware has been installed but be unable to identify the new hardware as a working sound card. To fix this problem, run the Setup software that came with the sound card.

- ✓ **Identifying CD-ROM/DVD Issues**

CD-ROM and DVD problems are normally media related. Although compact disc technology is much more reliable than that for floppy disks, it's not perfect. Another factor to consider is the cleanliness of the disc. On many occasions, if a disc is unreadable, cleaning it with an approved cleaner and a lint-free cleaning towel will fix the problem.

If the operating system doesn't see the drive, start troubleshooting by determining whether the drive is receiving power. If the tray will eject, you can assume there is power to it. Next, check BIOS Setup (for IDE drives) to make sure the drive has been detected. Once inside the case, ensure that the ribbon cable is properly aligned with pin 1 and that both the drive and motherboard ends are securely connected.

In order to play movies, a DVD drive must have MPEG decoding capability. This is usually accomplished via an expansion board, but it may be built into the video card or sound card, or it may require a software decoder. If DVD data discs will play but not movies, suspect a problem with the MPEG decoding.

If a CD-RW or DVD drive works normally as a regular CD-ROM drive but doesn't perform its special capability (doesn't read DVD discs or doesn't write to blank CDs), perhaps you need to install software to work with it. For example, with CD-RW drives, unless you're using an operating system such as Windows XP that supports CD writing, you must install CD-writing software in order to write to CDs.

- ✓ **Identifying NIC Issues**

In general, network interface cards (NICs) are added to a PC via an expansion slot. The most common issue that prevents network connectivity is a bad or unplugged patch cable.

Cleaning crews and the rollers on the bottoms of chairs are the most common threats to a patch cable. In most cases, wall jacks are placed 4 to 10 feet away from the desktop. The patch cables are normally lying exposed under the user's desk, and from time to time damage is done to the cable, or it's inadvertently snagged and unplugged. When you troubleshoot a network adapter, start with the most rudimentary explanations first. Make sure the patch cable is tightly plugged in, and then look at the card and see if any lights are on. If there are lights on, use the NIC's documentation to help troubleshoot. More often than not, shutting down the machine, unplugging the patch and power cables for a moment, and then reattaching them and rebooting the PC will fix an unresponsive NIC.

- ✓ **Identifying BIOS Issues**

Computer BIOSs don't go bad; they just become out-of-date. This isn't necessarily a critical issue they will continue to support the hardware that came with the box. It *does*, however, become an issue when the BIOS don't support some component that you would like to install larger hard drive, for instance.

Most of today's BIOSs are written to an EEPROM and can be updated through the use of software. Each manufacturer has its own method for accomplishing this. Check out the documentation for complete details.

- ✓ **Identifying Power-Supply Problems**

Power-supply problems can manifest themselves as a system that doesn't respond in any way when the power is turned on. When this happens, open the case, remove the power supply, and replace it with a new one. Partial failures, or intermittent power-supply problems, are much less simple. A completely failed power supply gives the same symptoms as a malfunctioning wall socket, UPS or power strip, a power cord that is not securely seated, or some motherboard shorts (such as those caused by an improperly seated expansion card, memory stick,
CPU, and the like), and you want to rule out those items before you replace the power supply and find you still have the same problem as when you started. Be aware that different cases have different types of on/off switches. The process of replacing a power supply is a lot easier if you purchase a replacement with the same mechanism.

- ✓ **Identifying Miscellaneous Problems**

Some common problems don't fit well into categories. This section lists some common hardware issues you'll be faced with.

✓ **Dislodged Chips and Cards**

The inside of a computer is a harsh environment. The temperature inside the case of some Pentium computers is well over 100F! When you turn on your computer, it heats up. Turn it off, and it cools down. After several hundred such cycles, some components can't handle the stress and begin to move out of their sockets. This phenomenon is known as *chip creep*, and it can be really frustrating.

Chip creep can affect any socketed device, including ICs, RAM chips, and expansion cards. The solution to chip creep is simple: Open the case, and reseat the devices. It's surprising how often this is the solution to phantom problems of all sorts.

Another important item worth mentioning is an unresponsive but freshly unboxed PC.

With the introduction of the Type II– and Type II–style of processors, the number of dead boxes increased dramatically. In fact, at that time I was leading a 2,000-unit migration for a large financial institution. As with any large migration, time and manpower were in short supply. The average dead PC ratio was about 1 out of every 20. When about 10 DOAs had stacked up, I stayed after work one night to assess the problem. After checking the power supply, RAM, and cables on these integrated systems, an examination of the chip provided me with the fix. These large, top-heavy processors can become dislodged during shipment.

Shortly after, manufacturers began using a heavier attachment point for the slot style of processor, which has helped tremendously.

✓ **Environmental Problems**

Computers are like human beings: They have similar tolerances to heat and cold. In general, anything comfortable to us is comfortable to computers. They need lots of clean, moving air to keep them functioning.

Dirt, grime, paint, smoke, and other airborne particles can become caked on the inside of the components. This is most common in automotive and manufacturing environments. The contaminants create a film that coats the components, causing them to overheat and/or conduct electricity on their surface. Blowing out these exposed systems with a can of condensed

One way to ensure that the environment has the least possible effect on your computer is to always leave the *blanks* in the empty slots on the back of your box. These pieces of metal are designed to keep dirt, dust, and other foreign matter from the inside of the computer. They also maintain proper airflow within the case to ensure that the computer doesn't overheat.